

Prompt Engineering

Knowledge-base document for the Domain-Specific RAG Chatbot

1. What is Prompt Engineering?

Prompt engineering is the practice of designing and refining instructions or inputs provided to a generative AI model so that the model is more likely to produce the desired type of output.

A useful prompt can specify the task, relevant context, output format, constraints, and other instructions needed for the application.

2. Components of a Useful Prompt

- Role or behavior: describes what the assistant should do.
- Task: clearly states what needs to be accomplished.
- Context: provides information that should be used to answer.
- Constraints: specifies what the model must or must not do.
- Output format: describes how the answer should be structured.

3. Zero-Shot Prompting

Zero-shot prompting asks the model to perform a task without providing worked examples of that task. The instruction describes the desired task directly.

For example, a prompt may ask a model to classify a sentence as positive or negative without showing example classifications.

4. Few-Shot Prompting

Few-shot prompting provides a small number of examples along with the task instruction. The examples demonstrate the desired relationship between inputs and outputs.

Few-shot examples can help a model follow a particular format or style without changing the model parameters.

5. Role Prompting

Role prompting assigns a behavior or perspective to the assistant, such as asking it to act as a technical documentation assistant. A role description can help organize the expected response behavior.

6. Structured Prompts

Structured prompts separate instructions, context, user questions, and output requirements. This is useful in RAG applications because retrieved document passages can be clearly separated from the user's question.

A RAG prompt can instruct the model to answer only from the retrieved context and to return a fallback message when the context does not contain the answer.

7. Hallucination and Guardrails

Prompt instructions can reduce unwanted behavior but cannot guarantee factual correctness by themselves. A strong RAG system combines prompt guardrails with retrieval, source metadata, testing, and appropriate model configuration.

A useful guardrail for a document assistant is: answer only from supplied context; do not invent facts; and state that the information could not be found when the context does not support an answer.

8. Prompt Engineering in This Project

The project uses a strict document-question-answering prompt. The retrieved chunks are supplied as context, and the model is instructed not to use outside knowledge. The response should also identify the source document and page when that metadata is available.